

Green Fn



Newton's Law of Cooling

Equation:

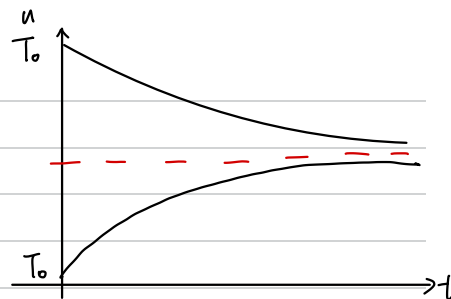
$$\frac{du}{dt} = -k(u - T_0)$$

transmission coefficient

Rate of temperature change $\propto$ Difference b/w ambient temperature and current temperature

Solution:

$$u(t) = T_0 + e^{-kt}$$



Fixed points (FP)

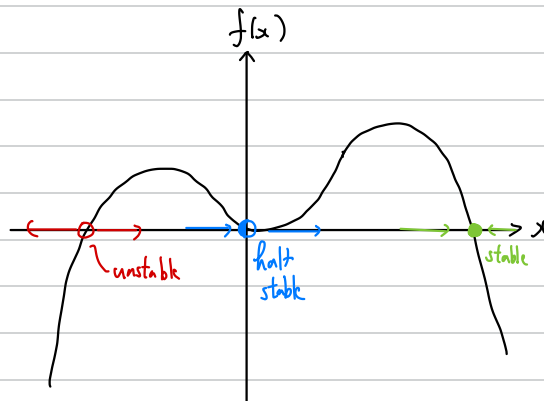
For $x' = f(x)$,

x_* is a fixed point if $f(x_*) = 0$

$$\left. \frac{df}{dx} \right|_{x=x_*} > 0 \rightarrow \text{unstable}$$

$$\left. \frac{df}{dx} \right|_{x=x_*} < 0 \rightarrow \text{stable}$$

$$\left. \frac{df}{dx} \right|_{x=x_*} = 0 \rightarrow \text{Further analysis req. (Possibly half stable)}$$



Linear Stability Analysis

$$x' = f(x)$$

$$\hookrightarrow \epsilon = x - x_*, \quad \epsilon \ll 1$$

$$\epsilon' = x' = f(\epsilon + x_*)$$

Taylor expansion = $f(x_*) + \left. \frac{df}{dx} \right|_{x=x_*} \cdot \epsilon + \frac{1}{2!} \left. \frac{d^2f}{dx^2} \right|_{x=x_*} \epsilon^2 + \dots$

$$= f(x_*) + \underbrace{\left. \frac{df}{dx} \right|_{x=x_*}}_A \cdot \epsilon$$

$$\text{Thus } \epsilon' = A\epsilon$$

$$\epsilon(t) = \epsilon_0 e^{At}$$

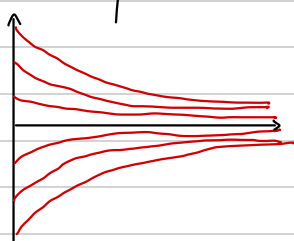
$$t \rightarrow \infty$$

$$\text{For } A > 0, \epsilon(t) \rightarrow \infty$$

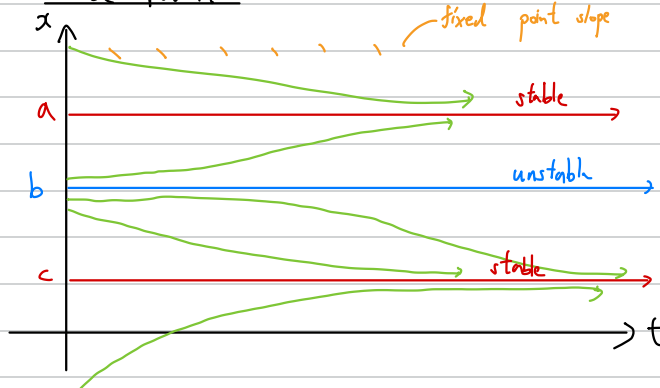
$$\text{For } A < 0, \epsilon(t) \rightarrow 0$$

Integral Curves

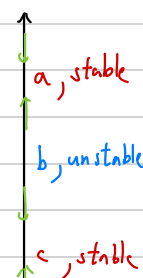
Infinite family of solutions



Phase Portrait



Phase Line



Principle of superposition, homogeneity

If $\{y_i\}_{i=1}^n$ are all solution to $a_0(t)y + a_n(t)y^{(n)} = 0$

then $\sum b_i y_i$ is also a solution

General Solution

$$y_g = A y_h + y_p$$

$$= \sum b_i y_{hi} + y_p$$

homogeneous particular

Separable Equations (Linear, separable)

It $\frac{dy}{dt} = f(t, y)$ can be written as

$$\frac{dy}{dt} = g(t) h(y) \quad \text{Separable, 1st Order}$$

$$\text{it } H(y) = \int \frac{dy}{h(y)}, \quad G(t) = \int g(t) dt$$

$$H(y) = G(t) + C$$

$$y = H^{-1}(G(t) + C)$$

Integrating factors (Linear)

$$y' + p(t)y = q(t) \quad \text{General Linear 1st Order}$$

IF, integrating factor, $\mu(t) = e^{\int p(t) dt}$

$$\text{Then } \mu y' + \underbrace{\mu p(t)}_{\mu'} y = \mu q(t)$$

$$y\mu = \int \mu q(t) dt$$

$$y = \underbrace{\exp(-\int p(t) dt)}_{1/\mu} \cdot \underbrace{\int \exp(\int p(t) dt) q(t) dt}_{\int \mu q dt}$$

Exact ODE (None Linear, Exact)

$$M(x, y) dx + N(x, y) dy = 0$$

Exact, 1st Order

$$\text{Exact when } M_y = \frac{\partial M}{\partial y} = N_x = \frac{\partial N}{\partial x},$$

$$\text{then } F(x, y) = C \text{ exists s.t. } F_x = M, F_y = N$$

Steps to solving an exact equation:

① Rearrange, ② Show exact, ③ Find $F(x, y) = C$

Bifurcation

Parameter $\longrightarrow a$

Point of bifurcation $\longrightarrow a = a$ s.t. FP changes

Dashed $\longrightarrow$ unstable, solid stable

Conditions for bifurcation

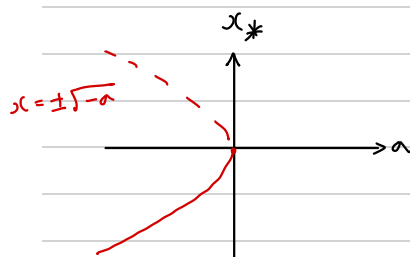
Suppose (x_*, r_c) is a candidate point for bifurcation

$$\hookrightarrow f(x_*, r_c) = 0$$

$$\hookrightarrow \frac{\partial f}{\partial x} \Big|_{x_*, r_c} = 0, \text{ Non-hyperbolicity}$$

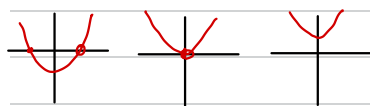
$$\hookrightarrow \frac{\partial f}{\partial r} \Big|_{x_*, r_c} \neq 0, \text{ Transversality}$$

Saddle node



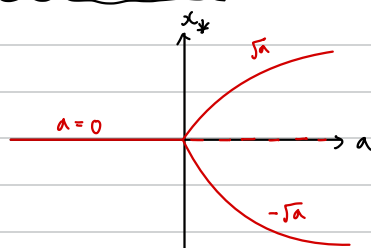
2FP $\longrightarrow$ 0FP

$a < 0$ $a = 0$ $a > 0$



$$f_{xx} \Big|_{x_*, r_c} \neq 0$$

Pitchfork Bifurcation



1FP @ 0 $\longrightarrow$ 3FP

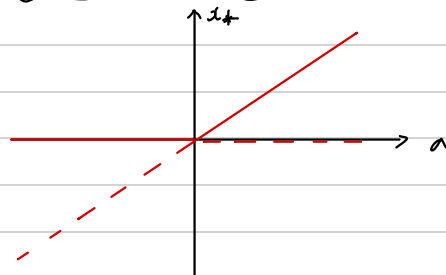
$a < 0$ $a = 0$ $a > 0$



$$f_{xx} \Big|_{x_*, r_c} = 0$$

$$f_{xxx} \Big|_{x_*, r_c} \neq 0$$

Transcritical Bifurcation



Stability changes

$a < 0$ $a = 0$ $a > 0$



Existence, Uniqueness of 1st order ODEs

Linear Case: $y' + p(t)y = q(t)$, $y(t_0) = y_0$

Existence: Consider the integrating factor method,

as long as p, q cont.,

$$y(t) = \frac{1}{\mu} \left[y_0 + \int_{t_0}^t \mu(s) q(s) ds \right] \text{ is well defined}$$

Uniqueness: Assume y_1, y_2 are sol^s, $y_1(t_0) = y_0 = y_2(t_0)$

Then let $z = y_1 - y_2$

$$\hookrightarrow z' + pz = 0$$

$$\hookrightarrow z(t_0) = 0$$

$$\text{Thus } \mu = \exp\left(\int p(t) dt\right)$$

$$(\mu z)' = 0$$

$$\mu z = C$$

But consider t_0, y_0

$$\mu(t_0) z(t_0) = C$$

$$\mu(t) z(t) = 0$$

$$\mu(t) = e^{\int p(t) dt} \neq 0$$

Hence

$$z(t) = 0$$

$$y_1 = y_2$$

General Case: Cauchy-Lipschitz:

$$\begin{cases} y' = f(t, y) \\ y(t_0) = y_0 \end{cases}$$

If $f, \frac{df}{dy}$ are cont. on $R = [\alpha, \beta] \times [\gamma, \delta]$ and $(t_0, y_0) \in \text{interior}(R)$ (excludes boundaries)

Then there exists a unique solⁿ on some time interval $[t_0 - h, t_0 + h]$, for some $h < \min\{|t_0 - \alpha|, |\beta - t_0|\}$

Note: Uniqueness depends on the continuity of $\frac{df}{dy}$

Population models

Simple model

$$\text{Eqn: } P' = rP$$

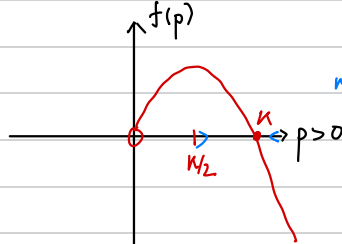
$$\text{Sol: } P = P_0 e^{rt}$$

$\hookrightarrow$ Non-realistic,
no ceiling

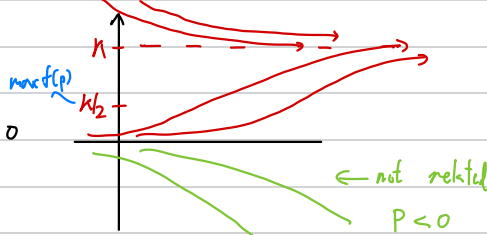
Logistic Model

$$\text{Eqn: } P' = rP \left(1 - \frac{P}{K}\right) \quad \text{carrying capacity}$$

Fixed Point:



Phase Portrait:



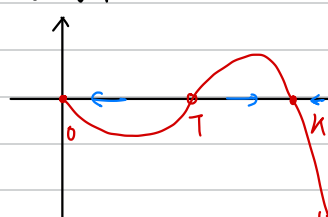
$$\text{Sol}^n: P(t) = \frac{K}{1 + \left(\frac{K - P_0}{P_0}\right) e^{-rt}}$$

Logistic w/ Threshold

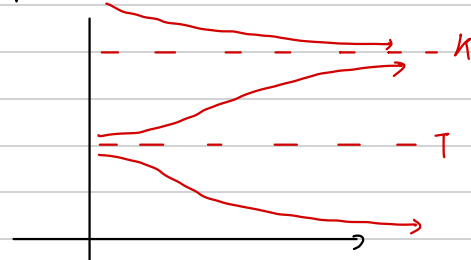
$$\text{Eqn: } P' = -rP \left(1 - \frac{P}{T}\right) \left(1 - \frac{P}{K}\right) \quad \text{Threshold}$$

$$\begin{cases} 0 < P_0 < T, P \rightarrow 0 \\ P_0 > K, P \rightarrow K \\ T < P_0 < K, P \rightarrow 0 \end{cases}$$

Fixed Point:



Phase Portrait:



Logistic in Harvesting

Eqn: $P' = rP \left(1 - \frac{P}{K}\right) - h$ constant harvesting rate

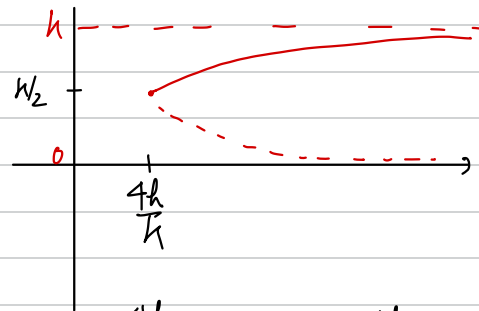
Fixed Points: $P_* = \frac{K}{2} \pm \frac{K}{2r} \sqrt{r(r - \frac{4h}{K})}$, $r, K > 0$

If $r - \frac{4h}{K} < 0 \rightarrow$ No FP

If $r - \frac{4h}{K} = 0 \rightarrow$ 1 FP, $P_* = \frac{K}{2}$

If $r - \frac{4h}{K} > 0 \rightarrow$ 2 FP

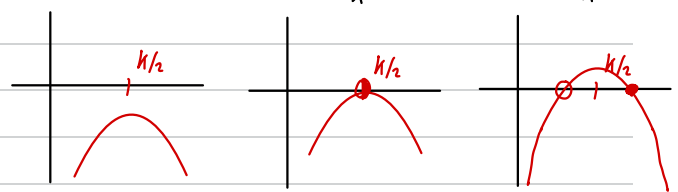
Saddle
node
Bifurcation



$r < \frac{4h}{K}$

$r = \frac{4h}{K}$

$r > \frac{4h}{K}$



2 species model

$$\begin{cases} x' = x(r_1 - ax - by) \\ y' = y(r_2 - cx - dy) \end{cases}$$

Where,

r_1, r_2 are intrinsic growth rates
 a, d are indiv. carrying capacity
 b, c are intra-species competition.

Lotka-Volterra

$$\begin{cases} x' = ax - bxy = x(a - by) \\ y' = -cy + dxy = y(-c + dx) \end{cases}$$

Assume:

x - prey a - growth rate of prey
 y - predator b - predation efficiency
c - predation death rate
d - Conversion efficiency
ie. Prey $\rightarrow$ Predator

Fixed-points: $(x_*, y_*) = (0, 0), (\frac{c}{d}, \frac{a}{b})$

$\hookrightarrow$ Nullclines:

$\hookrightarrow$ x-nullclines

$x = 0 \rightarrow \frac{dy}{dt} = -cy < 0$

or

$y = \frac{a}{b} \rightarrow \frac{dx}{dt} = \frac{a}{b} (dx - c) \begin{cases} > 0 \text{ for } x > \frac{c}{d} \\ < 0 \text{ for } x < \frac{c}{d} \end{cases}$

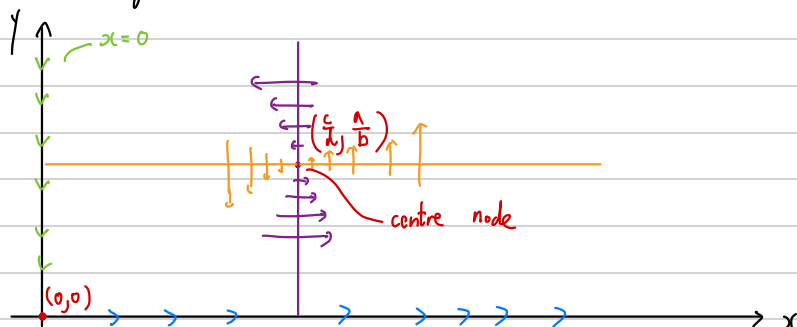
$\hookrightarrow$ y-nullclines:

$y = 0 \rightarrow \frac{dx}{dt} = ax > 0$

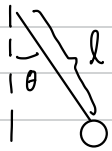
or

$x = \frac{c}{d} \rightarrow \frac{dy}{dt} = \frac{c}{d} (a - by) \begin{cases} > 0 \text{ for } y < \frac{a}{b} \\ < 0 \text{ for } y > \frac{a}{b} \end{cases}$

Phase diagram:

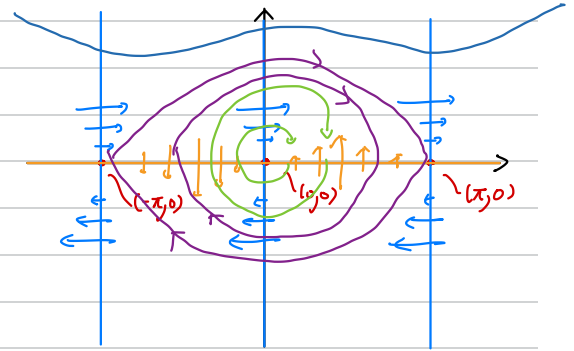


Simple Pendulum



Eqn': $\ddot{\theta} + \frac{g}{l} \sin \theta = 0 \rightarrow x' = y$
 If $x = \theta, y = \dot{\theta}$ $y' = -\frac{g}{l} \sin x$

Phase diagram:



Fixed points:

$$y = 0, \\ x = n\pi, n \in \mathbb{Z}$$

Nullclines

$\hookrightarrow x$ -nullcline:

$$y = 0$$

$$\frac{dy}{dt} = -\frac{g}{l} \sin x$$

$$\begin{cases} > 0, 0 < x < \pi \\ < 0, -\pi < x < 0 \end{cases}$$

$\hookrightarrow y$ -nullcline:

$$x = n\pi$$

$$\frac{dx}{dt} = y \begin{cases} > 0, y > 0 \\ < 0, y < 0 \end{cases}$$

Linear systems of Equations

General system:

$$x_1' = \sum_{i=1}^n P_{1i}(t) x_i + g_1(t)$$

$\vdots$

$$x_n' = \sum_{i=1}^n P_{ni}(t) x_i + g_n(t)$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}' = \begin{bmatrix} P_{11}(t) & P_{12}(t) & \dots & P_{1n}(t) \\ P_{21}(t) & & & \\ \vdots & & & \\ P_{n1}(t) & \dots & \dots & P_{nn}(t) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{bmatrix}$$

$X' = A X + B$

Homogeneity

If $X' = AX + \vec{b}$, then X_* must occur at $-A^{-1}\vec{b}$, A invertible,

Let $Y = X + A^{-1}\vec{b}$

$$Y' = X'$$

$$AY = A(X + A^{-1}\vec{b})$$

$$AY = AX + \vec{b}$$

$$= X' = Y$$

$$Y_* = \vec{0}$$

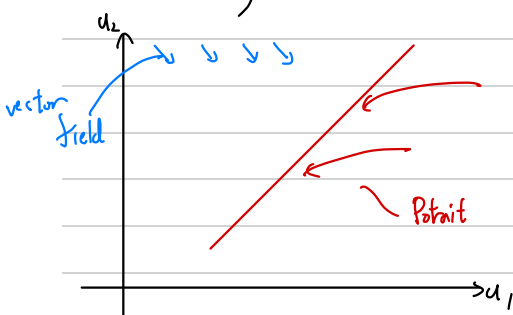
$\therefore X = Y - A^{-1}\vec{b}$, we can solve $Y' = AY$ first
 then shift by $-A^{-1}\vec{b}$

Side note: Nullclines

x -nullcline: When $x' = 0$

y -nullcline: When $y' = 0$

Direction Fields, Phase Portraits



Existence, Uniqueness

If $P_{ij}(t)$ and $g_i(t)$ are cont on (a, b) , continuing to for

all $i, j = 1, 2, \dots, n$ then IVP

$$X' = AX + \vec{b}, \quad X(t_0) = \vec{X}_0 \text{ has a unique sol. on } (a, b)$$

Homogenous, Linear, Constant coefficient systems

Principle of superposition

If X_1 and X_2 both satisfy $X' = AX$
then $c_1 X_1 + c_2 X_2$ is also a solⁿ

General solⁿ

n -dim: $X(t) = \sum_{i=1}^n c_i \vec{v}_i e^{\lambda_i t}$, $\{\lambda_i, \vec{v}_i\}_{i=1}^n$ are **distinct** eigenvalue, eigenvector pairs

2-dim: $X(t) = c_1 \vec{v}_1 e^{\lambda_1 t} + c_2 \vec{v}_2 e^{\lambda_2 t}$

2D Case

λ :

$$\lambda^2 - \text{tr} A \lambda + \det A = 0$$

$$\lambda = \frac{\text{tr} A \pm \sqrt{(\text{tr} A)^2 - 4 \det A}}{2}$$

$\vec{v}$: Solve $(A - \lambda I)\vec{v} = \vec{0}$

where $\det A = ad - bc$

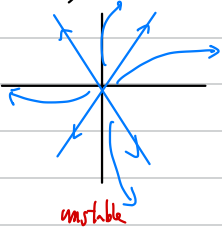
$$\text{tr} A = a + d$$

Real, Distinct Eigenvalues

i.e. $\lambda_1 \neq \lambda_2 \longrightarrow \text{tr} A^2 > 4 \det A$

Nodal source

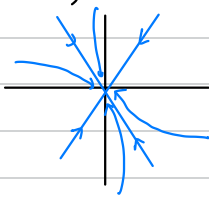
$$\lambda_1, \lambda_2 > 0$$



unstable

Nodal sink

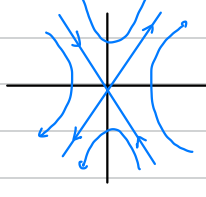
$$\lambda_1, \lambda_2 < 0$$



stable

Saddle node

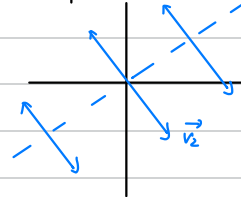
$$\lambda_1 > 0, \lambda_2 < 0 \text{ (or vice versa)}$$



stable

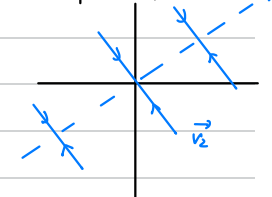
Line of critical points

$$\lambda_1 = 0, \lambda_2 > 0$$



unstable

$$\lambda_1 = 0, \lambda_2 < 0$$



stable

Infinitely many critical points along $\vec{v}_1$

If $\lambda_1 \neq 0$

$$x(t) = c_1 \vec{v}_1 e^{\lambda_1 t} + c_2 \vec{v}_2 e^{\lambda_2 t}$$

If $\lambda_1 = 0$,

$$x(t) = c_1 \vec{v}_1 + c_2 \vec{v}_2 e^{\lambda_2 t}$$

If $\lambda_1, \lambda_2 = 0$, any points are fixed, trivial system

Complex eigenvalues

$$\text{i.e. } \lambda_{1,2} = \alpha \pm i\beta$$

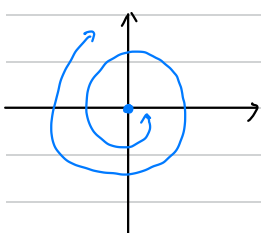
$$\vec{v}_{1,2} = \vec{p} \pm i\vec{q}$$

Conjugate pairs $\rightarrow$ only solve once

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Spiral Sink

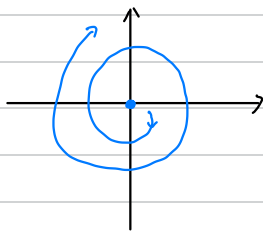
$$\lambda_{1,2} = \alpha \pm i\beta, \alpha < 0$$



stable

Spiral Source

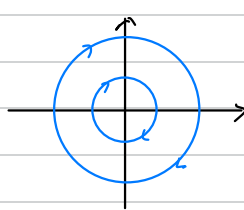
$$\lambda_{1,2} = \alpha \pm i\beta, \alpha > 0$$



unstable

Center

$$\lambda_{1,2} = \pm i\beta, \alpha = 0$$



$$\text{Sol}^n: x(t) = e^{\alpha t} [c_1 (\vec{p} \cos(\beta t) - \vec{q} \sin(\beta t)) + c_2 (\vec{p} \sin(\beta t) + \vec{q} \cos(\beta t))]$$

$\hookrightarrow$ Spins counterclockwise when $\beta > 0$
clockwise when $\beta < 0$

radial growth

ellipse

Repeated Eigenvalues
i.e. $\lambda_1 = \lambda_2$

Star nodes

When $A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$,

i.e. 2 linearly indep.

$\vec{v}_1, \vec{v}_2$

$x(t) = e^{\lambda t} (c_1 \vec{v}_1 + c_2 \vec{v}_2)$

Improper / Degenerate nodes

Only 1 linearly indep $\vec{v}_1$

① Find Generalized eigenvector:

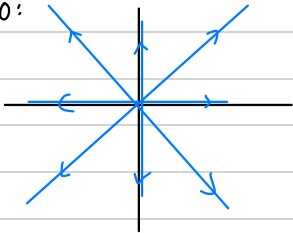
$(A - \lambda I) \vec{v}_2 = \vec{v}_1$

or $(A - \lambda I)^2 \vec{v}_2 = 0$

② General solⁿ:

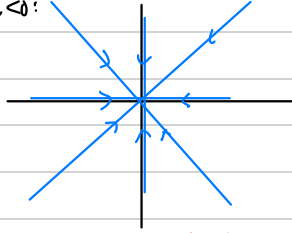
$x(t) = c_1 e^{\lambda t} \vec{v}_1 + c_2 e^{\lambda t} (t \vec{v}_1 + \vec{v}_2)$

$\lambda > 0$:



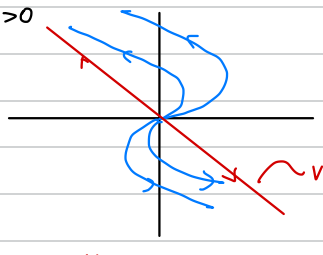
un stable star/proper node

$\lambda < 0$:



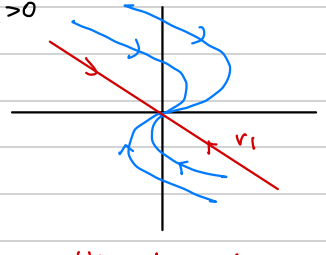
un stable star/proper node

$\lambda > 0$



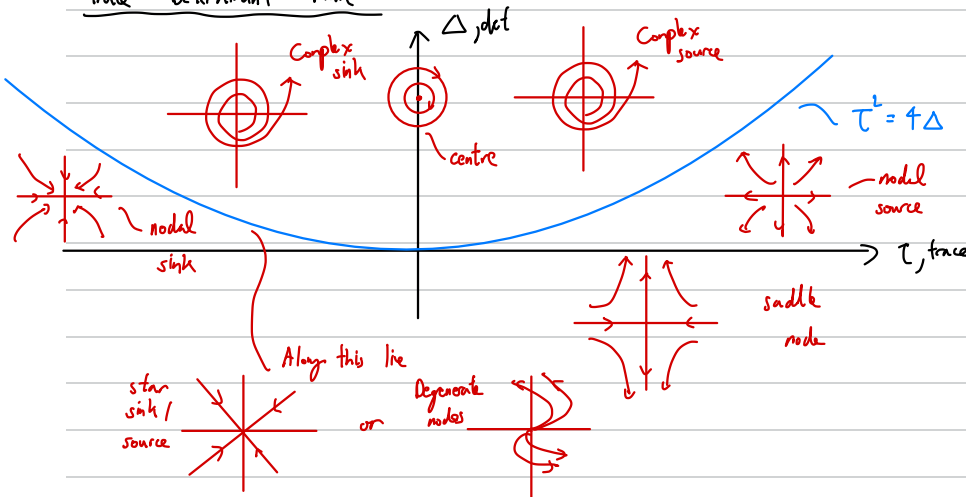
un stable degenerate/improper node

$\lambda < 0$



stable degenerate/improper node

Trace - Determinant Plane



Functionals

Maps a vector space of functions to real numbers

Wronskian

Functional,

$W[x_1, \dots, x_n](t) = \det \begin{pmatrix} x_1 & \dots & x_n \\ x_1' & \dots & x_n' \\ \vdots & \ddots & \vdots \\ x_1^{(n-1)} & \dots & x_n^{(n-1)} \end{pmatrix} (t) = 0$

$W(t) = 0$ everywhere iff $x_1, \dots, x_n$ are linearly dep.

$W(t) \neq 0$ everywhere iff $x_1, \dots, x_n$ are linearly indep.

Jacobian

For 2D:

$J_F[x, y] = \begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix}$ for $F = \begin{bmatrix} f(x, y) \\ g(x, y) \end{bmatrix}$

↳ We need n distinct sol's to describe the system

$W(t) = W(t_0) \exp \left(\int_{t_0}^t \text{tr}(P(s)) ds \right)$ for $X' = P(t)X$
 $W'(t) = \text{tr}(P(t)) W(t)$

LSA for 2D case

$$\begin{cases} x' = f(x, y) \\ y' = g(x, y) \end{cases}$$

Let x_*, y_* be FP

$$u = x - x_* \longrightarrow u' = x'$$

$$v = y - y_* \longrightarrow v' = y'$$

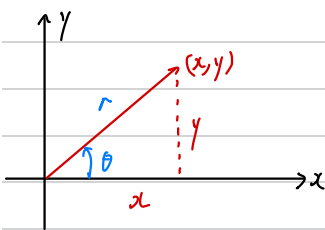
Now,

$$\begin{aligned} u' &= f(x, y) = f(x_*, y_*) + \frac{\partial f}{\partial x} \bigg|_{x_*, y_*} (u) + \frac{\partial f}{\partial y} \bigg|_{x_*, y_*} (v) + \theta(u^2, v^2) \dots \\ v' &= g(x, y) = g(x_*, y_*) + \frac{\partial g}{\partial x} \bigg|_{x_*, y_*} (u) + \frac{\partial g}{\partial y} \bigg|_{x_*, y_*} (v) + \phi(u^2, v^2) \dots \end{aligned} \longrightarrow \begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} \bigg|_{x_*, y_*} \begin{pmatrix} u \\ v \end{pmatrix}$$

$J(x, y)$

by assumption

Polar Coordinates



Conversions:

$$\begin{aligned} x &= r \cos \theta & \Rightarrow & \quad r^2 = x^2 + y^2 \\ y &= r \sin \theta & \quad \theta &= \arctan\left(\frac{y}{x}\right) \end{aligned}$$

Derivatives:

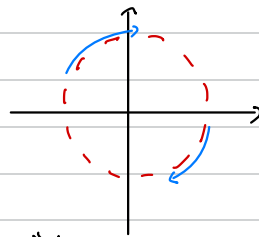
$$\begin{aligned} r' &= \frac{xx' + yy'}{r} = \frac{xx' + yy'}{\sqrt{x^2 + y^2}} \\ \theta' &= \frac{xy' - x'y}{x^2 + y^2} \end{aligned}$$

Notes:

$\theta' > 0 \rightarrow$ C.C.
 $\theta' < 0 \rightarrow$ C.
 Generally, $r > 0$

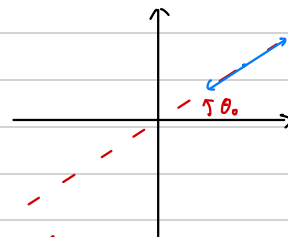
Nullclines:

$$r' = 0 \longrightarrow r(t) = r_0$$



"closed orbit"

$$\theta' = 0 \longrightarrow \theta(t) = \theta_0$$

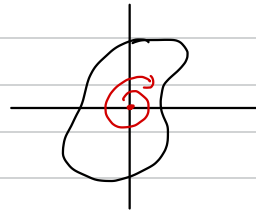


$$\frac{y'}{x'} = \frac{y}{x}$$

$\theta' = 0$ nullclines $\iff$ eigenvectors in Cartesian coordinates

Poincaré - Bendixson Theorem

In 2D dynamical systems, if trajectories that stay inside a bounded region which contains only, variable nodes, then all trajectories in this region must approach a limit cycle.



Steps for non-linear systems

① List equations

$$\begin{cases} x_1' = f_1(x_1, x_2, \dots, x_n) \\ \vdots \\ x_n' = f_n(x_1, x_2, \dots, x_n) \end{cases}$$

② Evaluate J at FP, find $\det(J|_{x_1^*, \dots, x_n^*})$,
 $\text{tr}(J|_{x_1^*, \dots, x_n^*})$

③ Consider $\text{tr} - \det$ Plane, find local behaviour.

④ Find the Jacobian

$$J(x_1, x_2, \dots, x_n) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

⑤ Repeat for Polar coordinates

Matrix Exponential

Scalar case: $e^t = \sum_{n=0}^{\infty} \frac{t^n}{n!}$

$n \times n$ case: $e^{At} = \sum_{n=0}^{\infty} \frac{A^n t^n}{n!}$

Diagonal Matrices

$$A = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix},$$

$$e^{At} = \sum_{k=0}^{\infty} \begin{pmatrix} \frac{\lambda_1^k t^k}{k!} & & 0 \\ & \ddots & \\ 0 & & \frac{\lambda_n^k t^k}{k!} \end{pmatrix}$$

$$e^{At} = \begin{pmatrix} e^{\lambda_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n t} \end{pmatrix}$$

Properties of e^{At}

1.) $e^{A \cdot 0} = I$

2.) $\frac{d}{dt} e^{At} = A e^{At}$

3.) $e^{A(s+t)} = e^{As} e^{At}$

4.) $(e^{At})^{-1} = e^{-At}$
inverse negative sign

5.) $e^{(A+B)t} = e^{At} e^{Bt}$ if A, B commute, i.e. $AB=BA$

Diagonalizable Matrices

$$A = V D V^{-1}, V = \begin{pmatrix} | & | & | \\ \vec{v}_1 & \vec{v}_2 & \dots & \vec{v}_n \\ | & | & | \end{pmatrix}, D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

Note $(V D V^{-1})^n = V D^n V^{-1}$

Then,
$$e^{At} = V \begin{pmatrix} e^{\lambda_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n t} \end{pmatrix} V^{-1}$$

Non-Diagonalizable Matrices

① Write $0 = \det(A - \lambda I)$

Find $\lambda_1, \vec{v}_1$

② Find $\vec{v}_j$ s.t. $(A - \lambda_1 I) \vec{v}_1 = \vec{0}$
 (if none are present, $(A - \lambda_1 I) \vec{v}_n = \vec{v}_{n-1}$)

③ $\vec{x}_1(t) = \vec{v}_1 e^{\lambda_1 t}$

$\vec{x}_2(t) = (\vec{v}_1 + t \vec{v}_1) e^{\lambda_1 t}$

$\vec{x}_3(t) = \left(\frac{t^2}{2} \vec{v}_1 + t \vec{v}_2 + \vec{v}_3 \right) e^{\lambda_1 t}$

$$X(t) = \begin{bmatrix} | & | & | \\ \vec{x}_1 & \vec{x}_2 & \dots & \vec{x}_n \\ | & | & | \end{bmatrix}$$

④ Since $e^{At} = X(t) X_0^{-1}$, we can pick $X_0 = X(0)$

↳ a. Find $X(0)$, b. find $X^{-1}(0)$, c. find $X(t) X^{-1}(0)$

Second Order Linear ODEs

$$y'' + p(t)y' + q(t)y = g(t)$$

Let $x_1 = y, x_2 = y'$

$x_1' = x_2, x_2' = -p(t)x_2 - q(t)x_1 + g(t)$

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -q(t) & -p(t) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ g(t) \end{bmatrix}$$

Existence, Uniqueness

Let $p(t), q(t), g(t)$ be cont. on an open interval I , let t_0 be any point in I , and $y_0, y_0' \in \mathbb{R}$, there exists a unique solution of $y = p(t)$ that satisfies $y'' + p(t)y' + q(t)y = g(t)$, and $y(t_0) = y_0, y'(t_0) = y_0'$, this solution exists throughout I

Homogeneous, Constant coefficients

$$ay'' + by' + cy = 0$$

$$X' = \begin{bmatrix} 0 & 1 \\ -c/a & -b/a \end{bmatrix} X$$

a) Distinct Real roots

$$\lambda_1, \vec{v}_1 = \begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix}$$

$$\lambda_2, \vec{v}_2 = \begin{pmatrix} 1 \\ \lambda_2 \end{pmatrix}$$

b) Repeated Real roots

$$\lambda_1, \vec{v}_1 = \begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix}$$

Generalized:

$$\vec{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

c) Complex conjugate roots

$$\lambda_{1,2} = \alpha \pm i\beta, \vec{v}_{1,2} = \begin{pmatrix} 1 \\ \lambda_{1,2} \end{pmatrix} = \begin{pmatrix} 1 \\ \alpha \pm i\beta \end{pmatrix}$$

$$\vec{x}_1 = e^{\alpha t} \begin{pmatrix} \cos \beta t \\ \alpha \cos \beta t - \beta \sin \beta t \end{pmatrix} + i e^{\alpha t} \begin{pmatrix} \sin \beta t \\ \alpha \sin \beta t + \beta \cos \beta t \end{pmatrix}$$

Thus $a\lambda^2 + b\lambda + c = 0$ $X = c_1 e^{\lambda_1 t} \begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix} + c_2 e^{\lambda_2 t} \begin{pmatrix} 1 \\ \lambda_2 \end{pmatrix}$ $X = c_1 e^{\lambda_1 t} \begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix} + c_2 e^{\lambda_1 t} \begin{pmatrix} t \\ 1 + \lambda_1 t \end{pmatrix}$ $\vec{X} = c_1 e^{\alpha t} \begin{pmatrix} \cos \beta t \\ \alpha \cos \beta t - \beta \sin \beta t \end{pmatrix} + c_2 e^{\alpha t} \begin{pmatrix} \sin \beta t \\ \alpha \sin \beta t + \beta \cos \beta t \end{pmatrix}$

Then $y = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$
 $y' = \lambda_1 c_1 e^{\lambda_1 t} + \lambda_2 c_2 e^{\lambda_2 t}$

$y = c_1 e^{\lambda_1 t} + c_2 t e^{\lambda_1 t}$
 $y' = \lambda_1 c_1 e^{\lambda_1 t} + (1 + \lambda_1 t) c_2 e^{\lambda_1 t}$

$y = c_1 e^{\alpha t} \cos \beta t + c_2 e^{\alpha t} \sin \beta t$
 $y' = c_1 e^{\alpha t} (\alpha \cos \beta t - \beta \sin \beta t) + c_2 e^{\alpha t} (\alpha \sin \beta t + \beta \cos \beta t)$

Cauchy Euler Equation

$$\text{Eqn: } ax'' + bx' + cy = 0$$

$$\text{Then guess } y = x^r, y' = rx^{r-1}, y'' = r(r-1)x^{r-2}$$

$$\text{Thus } a r(r-1)x^r + b r x^r + c x^r = 0$$

$$ar^2 - ar + br + c = 0$$

$$y(t) = c_1 x^{r_1} + c_2 x^{r_2}, \text{ for distinct } r$$

Operators

Second order differential operator

$$L = D^2 + pD + q \longrightarrow (D + \lambda_1 I)(D + \lambda_2 I)$$

Properties:

$$L[c_1 y_1 + c_2 y_2] = c_1 L[y_1] + c_2 L[y_2] \text{ (linearity)}$$

$$\text{If } y_1, y_2 \text{ solves } L[y] = 0, \text{ then } c_1 y_1 + c_2 y_2 \text{ also solves } L[y] = 0 \text{ (superposition)}$$

Homogeneous system operator

$$\text{Since } X' = AX$$

$$\text{then we define } K[x] = X' - P(t)X$$

Linearity

Superposition still holds.

Optional?

Eigenvalue, Function of linear operators

If $L[y]$ is an operator

λ is the eigenvalue, $y(t)$ is the eigenfunction that satisfies

$$L[y(t)] = \lambda y(t)$$

Derivative example

$$D[f] = \lambda f$$

$$\text{then } f = c e^{\lambda t}, \lambda \in \mathbb{C}$$

Fourier series

For a function where $y(0) = y(2\pi)$, we get

$$f(t) = e^{ikt}, k \in \mathbb{Z} \text{ as a basis,}$$

$$\text{then } f(t) \text{ can be expressed as: } \sum_{k=-\infty}^{\infty} c_k e^{ikt}, c_k = \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-ikt} dt$$

Generalized Eigenfunction

$$(L - \lambda I)^n [y(t)] = 0, I = \text{identity operator}$$

$L = \text{core operator}$

Applies to repeated eigenvalues

Non-homogeneity

Theorem: If y_1, y_2 satisfies the non-homogeneous equation,

then a) $y_1 - y_2$ satisfies the homogeneous equation,

b) $y_1 - y_2 = c_1 y_1 + c_2 y_2$, the fundamental set of sol.

$$\text{General sol'n: } y(t) = y_c(t) + y_p(t)$$

$$= c_1 y_1(t) + c_2 y_2(t) + y_p(t)$$

Constant coefficient: Undetermined coefficients

$$y'' + by' + cy = g(t)$$

Guess y_p takes the form of the following:

$g(t)$	$y_p(t)$
$P_n(t) = a_0 t^n + a_1 t^{n-1} + \dots + a_n$	$t^s (A_n t^n + A_{n-1} t^{n-1} + \dots + A_0)$
$P_n(t) e^{at}$	$t^s (A_n t^n + A_{n-1} t^{n-1} + \dots + A_0) e^{at}$
$P_n(t) e^{at} \begin{cases} \sin bt \\ \cos bt \end{cases}$	$t^s [(A_n t^n + A_{n-1} t^{n-1} + \dots + A_0) e^{at} \cos bt + (B_n t^n + B_{n-1} t^{n-1} + \dots + B_0) e^{at} \sin bt]$

t^s ensures no components solve the homogeneous equation

For 2nd order equations, $s = 0, 1$ or 2

Principle of Superposition in non-homogeneous solutions

If $G(t) = g_1(t) + g_2(t)$, and

$$y_{p1}(t) \text{ satisfies } ay'' + by' + cy = g_1(t),$$

$$y_{p2}(t) \text{ satisfies } ay'' + by' + cy = g_2(t),$$

$$\text{Then } Y_p(t) = y_{p1} + y_{p2} \text{ satisfies } ay'' + by' + cy = G(t)$$

Linear First Order Non-homogeneous 2-Dimensional system

$$\text{For } x' = P(t)x + G(t)$$

$$\vec{x}' = \begin{pmatrix} p_{11}(t) & p_{12}(t) \\ p_{21}(t) & p_{22}(t) \end{pmatrix} \vec{x} + \begin{pmatrix} g_1(t) \\ g_2(t) \end{pmatrix}$$

We assume a fundamental set of solution,

$$\vec{x}_1' = P(t)\vec{x}_1, \quad \vec{x}_2' = P(t)\vec{x}_2, \quad \text{note } W[\vec{x}_1, \vec{x}_2](t) \neq 0$$

Then the **fundamental matrix**,

$$X = \begin{pmatrix} \vec{x}_1 & \vec{x}_2 \end{pmatrix} \\ = \begin{pmatrix} x_{11}(t) & x_{12}(t) \\ x_{21}(t) & x_{22}(t) \end{pmatrix}$$

We guess $\vec{x}_p$ takes the form, $\vec{x}_p = u_1(t)\vec{x}_1(t) + u_2(t)\vec{x}_2(t) = X(t)u(t)$, $u(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix}$

we obtain

$$\star X(t) u'(t) = g(t)$$

$$u'(t) = X^{-1}(t)g(t), \quad \text{since } W[\vec{x}_1, \vec{x}_2] \neq 0 \quad \text{and} \quad X^{-1}(t) = \frac{1}{W[\vec{x}_1, \vec{x}_2]} \begin{pmatrix} x_{22}(t) & -x_{12}(t) \\ -x_{21}(t) & x_{11}(t) \end{pmatrix}$$

$$\star u(t) = \underbrace{\int X^{-1}(t)g(t) dt}_{\text{Particular solution}} + \underbrace{\vec{c}}_{\text{homogeneous solution}} = \underbrace{\begin{pmatrix} \int \frac{x_{22}g_1 - x_{12}g_2}{W} dt \\ \int \frac{x_{11}g_2 - x_{21}g_1}{W} dt \end{pmatrix}}_{\text{Particular solution}} + \underbrace{\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}}_{\text{Homogeneous portion}}$$

$$\text{Finally, } \vec{x} = \vec{x}_p + \vec{x}_c$$

$$= X(t) \left(\underbrace{\int X^{-1}(t)g(t) dt}_{\text{Particular}} + \underbrace{\vec{c}}_{\text{homogeneous}} \right)$$

Linear 2nd Order non-homogeneous, non-constant coefficients, differential equations

$$y'' + p(t)y' + q(t)y = g(t), \quad \text{Let } x_{11} = 0, x_{12} = 1, x_{21} = -q(t), x_{22} = -p(t), g_1 = 0, g_2 = g(t), X = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$x' = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} x + \begin{pmatrix} 0 \\ g(t) \end{pmatrix}, \quad y_1, y_2 \text{ be fundamental set,}$$

$$u_1(t) = - \int \frac{y_2(t)g(t)}{W[y_1, y_2]} dt + c_1, \quad u_2(t) = \int \frac{y_1(t)g(t)}{W[y_1, y_2]} dt + c_2$$

$$y = y_c + y_p \\ = c_1 y_1 + c_2 y_2 + \left(-y_1 \int \frac{y_2 g}{W} dt + y_2 \int \frac{y_1 g}{W} dt \right)$$

Reduction of order

Method of finding fundamental set, given y_1

Given y_1 , assume $y_2(t) = v(t)y_1(t)$, sub into original ODE, we obtain:

$$v''y_1 + v'(2y_1' + py_1) = 0$$

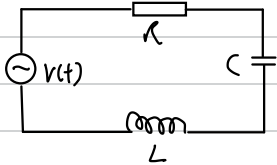
Thus

$$v(t) = \int v' dt = \int \exp\left(-\int \frac{2y_1' + py_1}{y_1} dt\right) dt$$

and $y_2 = v(t)y_1(t)$

Electrical Circuits

For a series RLC circuit, forcing component $V(t)$



$$V_L = L \frac{dq}{dt}, \quad V_C = \frac{q}{C}, \quad V_R = R \frac{dq}{dt}$$

Fundamental set:

$$\sum_i V_i = 0 \longrightarrow \begin{bmatrix} 0 & 1 \\ -1/C & -R/L \end{bmatrix} \begin{bmatrix} q \\ q' \end{bmatrix}$$

Autonomous

$$\frac{dy}{dt} = f(y)$$

Numerical Methods

$$\begin{cases} \frac{dy}{dt} = f(t, y) \\ y(t_0) = y_0 \end{cases}$$

↳ Euler's Method:

$$y_{n+1} = y_n + h f(t_n, y_n)$$

$$\text{LTE}, e_n \sim O(h^1)$$

$$|e_n| \leq \frac{M h^2}{2}, \text{ where } M \text{ is maximum of } f'(y, t) \text{ in interval.}$$

$$\text{GTE}, E_n \sim O(h)$$

$$E_n = \frac{n M h^2}{2}, n = n \text{ steps}$$

$$= \frac{T - t_0}{h} \cdot \frac{M h^2}{2}$$

$$= \frac{(T - t_0) M h}{2}$$

↳ Improved Euler:

$$\widetilde{y}_{n+1} = y_n + h f(t_n, y_n)$$

$$y_{n+1} = y_n + \frac{f(t_{n+1}, \widetilde{y}_{n+1}) + f(t_n, y_n)}{2} h$$

$$= y_n + \frac{f\left[t_n + h, y_n + h f(t_n, y_n)\right] + f(t_n, y_n)}{2} h$$

$$\text{LTE} \sim O(h^3)$$

$$\text{GTE} \sim O(h^2)$$

↳ Runge-Kutta

$$\text{RK2: } k_1 = f(t_n, y_n)$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right)$$

$$y_{n+1} = y_n + h k_2$$

$$\text{LTE} \sim O(h^3)$$

$$\text{GTE} \sim O(h^2)$$

$$\text{RK4: } k_1 = f(t_n, y_n)$$

$$k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_1\right)$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_2\right)$$

$$k_4 = f(t_n + h, y_n + h k_3)$$

$$y_{n+1} = y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$\text{LTE} \sim O(h^5)$$

$$\text{GTE} \sim O(h^4)$$

Generally,

$$k_1 = f(t_n, y_n)$$

$$k_2 = f(t_n + c_2 h, y_n + h a_{21} k_1)$$

⋮

$$k_s = f(t_n + c_s h, y_n + h \sum_j a_{sj} k_j)$$

$$\hookrightarrow y_{n+1} = y_n + h \sum_{i=1}^s b_i k_i$$

Lipschitz Continuity

$$|f(t, y_1) - f(t, y_2)| \leq L |y_1 - y_2|$$

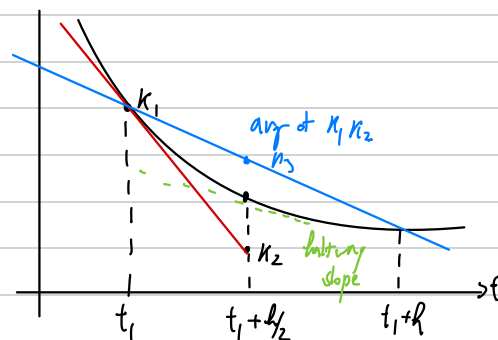
$y' = f(t, y)$, if f is Lip. cont., then

$$|E_n| \leq \frac{\max\{e_n\}}{hL} (e^{L(T-t_0)} - 1)$$

Theorem

If $e_n \sim O(h^p)$

then $E_n \sim O(h^{p-1})$



Laplace transformation

Definition: $L[f](s) = \int_0^{\infty} e^{-st} f(t) dt = F(s)$

Existence: $L[f](s)$ exists $\forall s$ if

① f is piecewise cont $\forall t \leq A$

② f is of exponential order, i.e.

there exists real constants $M \geq 0, K > 0$ and a s.t. $|f(t)| \leq M e^{at}$ when $t \geq M$

Corollary: if f satisfies ①, ②, $|F(s)| \leq L/s$ for some L ,
 $\lim_{s \rightarrow \infty} F(s) = 0$

Properties

↳ Linearity, if $L[f](s)$ and $L[g](s)$ exists for $f: \mathcal{R}_c\{s\} \geq a_1, g: \mathcal{R}_c\{s\} \geq a_2$, respectively
then $L[c_1 f + c_2 g](s) = c_1 L[f](s) + c_2 L[g](s)$ exist $\forall s$ s.t. $\mathcal{R}_c\{s\} \geq \max(a_1, a_2)$

↳ Derivatives, if ① $f, f', \dots, f^{(n-1)}$ are cont. and $f^{(n)}$ is piecewise cont. on $0 \leq t \leq A$
② $f, f', \dots, f^{(n-1)}, f^{(n)}$ are of exponential order,
then for $s > a$,

$$L\{f^{(n)}(t)\} = s^n L\{f(t)\} - s^{n-1} f(0) - \dots - s f^{(n-2)}(0) - f^{(n-1)}(0)$$

ex. $L\{f'(t)\} = s L\{f(t)\} - f(0)$ or $L\{f''(t)\} = s^2 L\{f(t)\} - s f(0) - f'(0)$

↳ Scale by e^{ct} , if $L\{f(t)\} = F(s)$ exists,

$$L\{e^{ct} f(t)\} = F(s-c), s > a+c$$

$$L\{t^n f(t)\} = (-1)^n F^{(n)}(s)$$

$$L\{t^n\} = n! / s^{n+1}$$

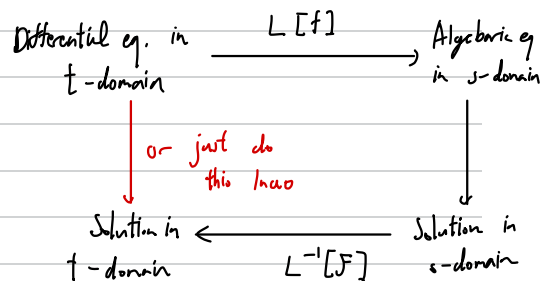
Inverse Laplace

Theorem, if ① f, g are piecewise cont. and of exponential order

② $F = G$, where $F = L\{f\}, G = L\{g\}$

then $f = g$ for all points f, g are cont.

↳ $f = L^{-1}\{F\}$, definition not relevant.



Unit Step / Heaviside Function and Indicator Function

Heaviside: $u_c(t) = \begin{cases} 1, & t \geq c \\ 0, & t < c \end{cases}$



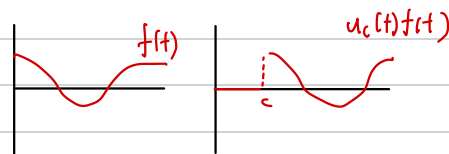
Indicator: $u_{cd}(t) = u_c(t) - u_d(t) = \begin{cases} 1, & c \leq t < d \\ 0, & t < c, t \geq d \end{cases}$



Laplace Transformed,
 $\hookrightarrow \mathcal{L}\{u_c(t)\} = \frac{e^{-cs}}{s}, s > 0$

$\hookrightarrow \mathcal{L}\{u_{cd}(t)\} = \frac{e^{-cs} - e^{-ds}}{s}$

$\hookrightarrow$ If $\mathcal{L}\{f(t)\}$ exists for $s > a \geq 0$, c is non-negative,
 $\mathcal{L}\{u_c(t) f(t-c)\} = e^{-cs} \mathcal{L}\{f(t)\}$



or

$u_c(t) f(t-c) = \mathcal{L}^{-1}\{e^{-cs} F(s)\}$

Periodic Functions

$\hookrightarrow f(t+T) = f(t)$ for $T > 0$

Window fun: $f_T(t) = f(t)[1 - u_T(t)] = \begin{cases} f(t), & 0 \leq t < T \\ 0, & \text{else.} \end{cases}$

$F_T(s) = \int_0^T e^{-st} f(t) dt$

Window fun shifted right
 $f_T(t - kT) u_{kT}(t) = \begin{cases} f(t - kT), & kT \leq t < (k+1)T \\ 0, & \text{else} \end{cases}$

$f(t) = \sum_{n=0}^{\infty} f_T(t - nT) u_{nT}(t)$

$\hookrightarrow$ If ① f is periodic with period T ,
 ② Piecewise cont. on $[0, T]$, then

$\mathcal{L}\{f(t)\} = \frac{F_T(s)}{1 - e^{-sT}} = \frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}},$ where

Dirac-Delta function

$\delta(t) = \begin{cases} 0, & t \neq 0 \\ \infty, & t = 0 \end{cases}$

$\hookrightarrow$ Properties:

$\int_{-\infty}^{\infty} \delta(t) dt = 1$

$\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}$

$\int_{-\infty}^{\infty} f(t) \delta(t - a) dt = f(a)$

$\mathcal{L}\{\delta(t)\} = 1$

Convolution

$$f * g(t) = \int_0^t f(t-\tau)g(\tau) d\tau$$

Properties:

$$\hookrightarrow f * g = g * f$$

$$f * \delta = f$$

$$\hookrightarrow f * (g_1 + g_2) = f * g_1 + f * g_2$$

$$\hookrightarrow (f * g) * h = f * (g * h)$$

$$\hookrightarrow f * 0 = 0 * f = 0$$

Theorem: if $L\{f(t)\} = F(s)$ and $L\{g(t)\} = G(s)$ both exist for $s > a \geq 0$,
 $H(s) = F(s)G(s) = L\{h(t)\}$

$$h(t) = \int_0^t f(\tau)g(t-\tau) d\tau \quad \text{or} \quad \int_0^t f(t-\tau)g(\tau) d\tau$$

Free, Forced response

For $ay'' + by' + cy = g(t)$, $y(0) = y_0$, $y'(0) = y_1$, where $H(s) = \frac{1}{as^2 + bs + c}$ (transfer function)

$$s: Y(s) = H(s) [(as + b)y_0 + ay_1] + H(s)G(s)$$

$$t: y(t) = \underbrace{L^{-1}[H(s) [(as + b)y_0 + ay_1]]}_{\text{free response}} + \underbrace{\int_0^t f(t-\tau)g(\tau) d\tau}_{\text{forced response}}$$

Fourier Series

$\hookrightarrow$ Decompose periodic function as a sum of sinusoids

$$\omega_k = \frac{2\pi k}{T}$$

Orthogonal Families

$\hookrightarrow$ Let $PC[a, b]$ be space of piecewise continuous f_s on $[a, b] \in \mathbb{R}$
piecewise cont.

$$PC[a, b] \supset C^0[a, b]$$

$\supset$

$\vdots$

$$PC[a, b] \supset C^1[a, b]$$

$$C^0 \supset C^1 \supset C^2 \dots \supset C^n[a, b]$$

$PC[a, b]$ is a vector space

$\hookrightarrow f, g \in PC$, then $\alpha f + g \in PC$, $\alpha \in \mathbb{R}$

Inner Products on PC [a,b]

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx, \quad f, g: \mathbb{R} \rightarrow \mathbb{R}$$

$$\hookrightarrow \langle f, g \rangle = \langle g, f \rangle$$

$$\hookrightarrow \langle f + \alpha g, h \rangle = \langle f, h \rangle + \alpha \langle g, h \rangle$$

$$\hookrightarrow \langle f, f \rangle \geq 0, \text{ Positive-definiteness}$$

$$\langle f, f \rangle = 0 \text{ if } f = 0$$

Orthogonality

f, g are orthogonal if $\langle f, g \rangle = 0$

Orthonormality

$$\|f\| = \sqrt{\langle f, f \rangle} \geq 0$$

$$\text{Orthonormal if } \|f\| = \|g\| = 1$$

$$\text{and } \langle f, g \rangle = 0$$

$\mathbb{R}^n$ Euclidean

Elements $v \in \mathbb{R}^n$

Dim n

$$\text{Inner prod. } \langle v, w \rangle = \sum_{i=1}^n v_i w_i$$

ONB $\{v_i\}$ s.t. $\langle v_i, v_j \rangle = \delta_{ij}$

$$\text{Span } W = \sum a_i v_i$$

PC $[-L, L]$

$f(x)$

∞

$$\langle f, g \rangle = \int_{-L}^L f(x)g(x) dx$$

$$\left\{ 1, \sin\left(\frac{n\pi x}{L}\right), \cos\left(\frac{n\pi x}{L}\right) \right\}$$

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)$$

Fourier's Theorem

Suppose f is periodic, with period $2L$, and $f, f' \in PC[-L, L]$

Then

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

Orthonormal basis:

$$\left\langle \sqrt{\frac{2}{L}} \cdot \frac{1}{2}, \frac{1}{\sqrt{L}} \sin\left(\frac{n\pi x}{L}\right), \frac{1}{\sqrt{L}} \cos\left(\frac{n\pi x}{L}\right) \right\rangle$$

$$a_n = \left\langle f, \frac{1}{\sqrt{L}} \cos\left(\frac{n\pi x}{L}\right) \right\rangle = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \left\langle f, \frac{1}{\sqrt{L}} \sin\left(\frac{n\pi x}{L}\right) \right\rangle = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$a_0 = \left\langle f, \sqrt{\frac{2}{L}} \cdot \frac{1}{2} \right\rangle = \frac{1}{L} \int_{-L}^L f(x) dx$$

PDEs

1D. Heat equation.

$$\begin{cases} \frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2} \\ u(x, 0) = f(x) \text{ , initial cond.} \\ u(0, t) = u(L, t) = 0 \text{ , boundary cond.} \end{cases}$$

Assume separable, $u(x, t) = X(x) T(t)$

Conditions give:

$$u(x, 0) = f(x) = X(x) T(0)$$

$$u(0, t) = X(0) T(t) = 0, \quad X(0) = 0$$

$$u(L, t) = X(L) T(t) = 0, \quad X(L) = 0$$

$$X(x) T'(t) = \alpha^2 X''(x) T(t)$$

$$\frac{1}{\alpha^2} \frac{T'}{T} = \frac{X''}{X} = -\lambda \quad \text{convention}$$

$$\begin{array}{cc} \swarrow & \searrow \\ X'' = -\lambda X & T' = -\alpha^2 \lambda T \end{array}$$

$$\textcircled{1} \quad X(x) = A \cos(\sqrt{\lambda} x) + B \sin(\sqrt{\lambda} x) \quad \textcircled{2} \quad T(t) = e^{-\alpha^2 \lambda t}$$

$$X(0) = 0 \rightarrow A = 0$$

$$T_n(t) = \exp\left(-\frac{\alpha^2 n^2 \pi^2}{L^2} t\right)$$

$$X(x) = B \sin(\sqrt{\lambda} x)$$

$$X(L) = 0 \rightarrow \lambda \pi = \sqrt{\lambda} L$$

$$\lambda = \frac{n^2 \pi^2}{L^2}$$

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

$$\textcircled{3} \quad u_n(x, t) = \sum_0^\infty c_n \exp\left(-\frac{\alpha^2 n^2 \pi^2}{L^2} t\right) \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = u(x, 0) = \sum_0^\infty c_n \sin\left(\frac{n\pi x}{L}\right)$$

$$c_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \quad (\text{Fourier coeff})$$

Fourier Transform

$$f \in L^p(\mathbb{R}) \rightarrow \left(\int |f|^p dx \right)^{1/p} < \infty$$

$$\hookrightarrow \mathcal{F}[f](\omega) = \hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx, \text{ defined for } f \in L^1(\mathbb{R}) \rightarrow \lim_{\omega \rightarrow \pm\infty} |\hat{f}(\omega)| = 0$$

$$\mathcal{F}^{-1}[\hat{f}](x) = f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega$$

Fourier Series $\rightarrow$ Transform

periodic case $\rightarrow$ Discrete set of frequencies

Non-periodic $\rightarrow$ Continuous set of freq.

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in x} \rightarrow \cos(nx) + i \sin(nx)$$

$$\hookrightarrow [-L, L]$$

$$\hookrightarrow c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$\hookrightarrow c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-n\pi x/L} dx$$

$$\text{Then as } L \rightarrow \infty, \omega_n = \frac{n\pi}{L}, \Delta\omega = \frac{\pi}{L} \rightarrow 0$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega$$

"coefficient"

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

Properties of Fourier Transform

① Linearity, $\widehat{af+g} = a\hat{f} + \hat{g}$

② Derivative, $\mathcal{F}\left[\frac{df}{dx}\right]$

$$\mathcal{F}[f^{(n)}](\omega) = (i\omega)^n \hat{f}(\omega)$$

$$(\hat{f}' \neq \hat{f}') \quad \hat{f}' = \int_{-\infty}^{\infty} \frac{df}{dx} e^{-i\omega x} dx$$
$$= i\omega \hat{f}(\omega)$$

⑤ Convolutions

$$\mathcal{F}[f * g](\omega) = \hat{f} \cdot \hat{g}$$

⑥ Translations

$$\widehat{f(x-a)} = e^{-i\omega a} \hat{f}(\omega)$$

Fourier Transform and functions

$$\mathcal{F}[1](\omega) = 2\pi \delta(\omega)$$

$$\int_{-\infty}^{\infty} e^{-i\omega x} dx = 2\pi \delta(\omega)$$

$$\mathcal{F}[\sin ax] = -\pi i [\delta(\omega - a) - \delta(\omega + a)] \quad \text{compared to } \mathcal{L}[f] = \frac{a}{s^2 + a^2} \rightarrow s = \pm ia \text{ (extinct freq)}$$

Fourier and heat eq

$$\begin{cases} u_t = \alpha u_{xx} \\ u(x, 0) = f(x) \end{cases}$$

$$\hat{u}_t = -\alpha \omega^2 \hat{u}$$

$$\hookrightarrow \hat{u}(\omega, t) = \hat{u}(\omega, 0) \exp(-\alpha \omega^2 t)$$

$$= \hat{f}(\omega) \exp(-\alpha \omega^2 t)$$

$$u(x, t) = \mathcal{F}^{-1}[\hat{u}(\omega, t)](x)$$

$$= \mathcal{F}^{-1}[\underbrace{\hat{f}(\omega)}_{\mathcal{F}[f]} \cdot \underbrace{\exp(-\alpha \omega^2 t)}_{\hat{G}(\omega, t) = e^{-\alpha \omega^2 t}}](x)$$

$$= f * G(x, t)$$

$$G(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\alpha \omega^2 t} e^{-i\omega x} d\omega$$
$$= \frac{1}{\sqrt{4\pi\alpha t}} e^{-\frac{x^2}{4\alpha t}}$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi\alpha t}} e^{-\frac{(x-y)^2}{4\alpha t}} f(y) dy$$

Dirichlet Problem on the Disk

$$\begin{cases} \Delta u = u_{xx} + u_{yy} = 0, & x^2 + y^2 < a^2 \\ u(x, y) = h(\theta), & x^2 + y^2 = a^2 \end{cases}$$

Polar coords

$$\hookrightarrow R(r) \Theta(\theta)$$

$$\frac{\partial}{\partial x} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}$$

$$\frac{\partial}{\partial y} = \sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta}$$

$$\Delta u = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} = \Delta_{r, \theta}$$

$$\text{Now } \begin{cases} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, & r < a \\ u(a, \theta) = h(\theta), & r = a \end{cases}$$

Assuming $u = R(r) \Theta(\theta)$,

$$0 = \underbrace{\frac{r^2 R''}{R}}_{+1} + \underbrace{\frac{r R'}{R}}_{-1} + \underbrace{\frac{\Theta''}{\Theta}}_{-1}$$

$$\frac{\Theta''}{\Theta} = -1$$

$$\Theta'' = -1 \cdot \Theta, \quad \Theta(\theta) = A \cos(\sqrt{1} \theta) + B \sin(\sqrt{1} \theta)$$

$$\sqrt{1} \sim n$$

$$1_n = n^2$$

$$\Theta_n(\theta) = A_n \cos(n\theta) + B_n \sin(n\theta)$$

$$\frac{r^2 R''}{R} + \frac{r R'}{R} = 1 \rightarrow \text{Cauchy-Euler}$$

$$R(r) = r^l$$

$$[l(l-1) + l - 1] r^l \rightarrow l = \pm n$$

$$R_n(r) = C r^n + D r^{-n}$$

$$R_0(r) = C + D \log r$$

$$R_n(r) = r^n$$

Then $u_n(r, \theta) = r^n (A_n \cos(n\theta) + B_n \sin(n\theta))$

$$u(r, \theta) = \sum_{n=0}^{\infty} u_n(r, \theta)$$

$$= \frac{A_0}{2} + \sum_{n=1}^{\infty} r^n [A_n \cos(n\theta) + B_n \sin(n\theta)]$$

$$u(r, \theta) = f(\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} a^n [A_n \cos(n\theta) + B_n \sin(n\theta)]$$

$$A_n = \frac{1}{\pi a^n} \int_{-\pi}^{\pi} f(\theta) \cos(n\theta) d\theta$$

$$B_n = \frac{1}{\pi a^n} \int_{-\pi}^{\pi} f(\theta) \sin(n\theta) d\theta$$